

1. Overview

The **UNT MAP 2D Project** is an indoor navigation system that combines a Unity-based application with a React web interface. The Unity component handles navigation logic and pathfinding, while the React website serves as a front-end to display the Unity WebGL build. Additionally, an Electron setup allows the application to run as a desktop app.

The purpose of this document is to describe how to deploy, configure, and run the system across development and production environments.

2. Assumptions

- Users have **Unity Hub** installed with version **2022.3.62f3**
 - Users have **Node.js** and **npm** installed
 - Git is installed and configured
 - The system has internet access to install dependencies
 - Users have basic knowledge of:
 - Command line usage
 - React/Node project setup
 - Unity project handling
 - The WebGL build is already generated in Unity before deployment
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3. Dependencies

Core Dependencies

- Unity Editor **2022.3.62f3**
- Node.js (v16+ recommended)
- npm (comes with Node)

Frontend (React Website)

- Vite (development server)
- React (latest stable version)
- Electron (for desktop app support)

Build Tools

- esbuild (version 0.17.19 if issues occur)

External Components

- Unity WebGL build (exported from Unity)
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4. Constraints

- WebGL builds may have **performance limitations** depending on the browser
 - The system requires a **modern browser** (Chrome, Edge, etc.)
 - Electron apps may behave differently across operating systems
 - macOS users may encounter **esbuild compatibility issues**
 - Unity scene files cannot be edited simultaneously without merge conflicts
 - The project must maintain the correct folder structure for WebGL integration
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5. Description of Deployment Artifacts

Project Structure

UNT-MAP-2D/

```
| — Assets/           # Unity assets and scenes
| — Packages/         # Unity package dependencies
| — ProjectSettings/  # Unity configuration files
| — Sprints/          # Project planning files
| — unt-map-website/  # React + Electron application
| — README.md
```

Website Structure

unt-map-website/

```
| — public/
|   | — map/           # Unity WebGL build files
|   |   | — Build/
|   |   | — TemplateData/
|   |   | — index.html
| — src/               # React source code
| — package.json       # Dependencies and scripts
```

Artifact Roles

- **Unity Project** → Handles navigation logic and generates WebGL build
 - **WebGL Build** → Deployed output used by the website
 - **React App** → Displays the WebGL content in browser
 - **Electron App** → Wraps the website into a desktop application
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6. Data Creation

No traditional database setup is required.

However, the following initialization steps are necessary:

- Build the Unity project as a **WebGL build**
- Copy the generated files into:

unt-map-website/public/map/

- Ensure the following exist:
 - Build/
 - TemplateData/
 - index.html

This acts as the “data initialization” for the system.

7. Admin Credentials

- No admin login system is currently implemented in this project
 - If added in the future, credentials should be stored securely using environment variables
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8. Deployment Process

Step 1: Clone Repository

```
git clone <repository-url>  
cd UNT-MAP-2D
```

Step 2: Open Unity Project

- Open **Unity Hub**
 - Select the UNT-MAP-2D folder
 - Ensure Unity version **2022.3.62f3** is used
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Step 3: Build WebGL (Unity)

- In Unity:
 - Go to **File** → **Build Settings**
 - Select **WebGL**
 - Click **Build**
 - Output the build files
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Step 4: Move WebGL Build

Copy the build output into:

```
unt-map-website/public/map/
```

Step 5: Setup Website

```
cd unt-map-website  
npm install  
npm run dev
```

Open:

<http://localhost:5173>

Step 6: Run Electron App (Optional)

`npm run electron:dev`

To build:

`npm run electron:build`

Step 7: Fix macOS Issues (if needed)

`rm -rf node_modules package-lock.json`

`npm install esbuild@0.17.19`

`npm install`

Step 8: Verify Deployment

- Website loads successfully
- Unity WebGL map renders correctly
- Navigation features function as expected
- Electron app launches (if used)